# Supplementary Online Content

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This supplementary material has been provided by the authors to give readers additional information about their work.

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**eAppendix.** Search Strategies

We searched PubMed, EMBASE, and the Cochrane Register of Controlled Trials (CENTRAL) in The Cochrane Library, on March 5, 2020. The search strategies were listed below.

**PubMed**

#1 Migraine Disorders[Mesh] OR Migrain*[tiab]

#2 Drug Therapy[Mesh] OR Drug*[tiab] OR Chemotherap*[tiab] OR Pharma*[Mesh] OR Therapeutics[Mesh] OR Therap*[tiab] OR Treat*[tiab] OR Treatment Outcome[Mesh] OR Outcome*[tiab]

#3 #1 AND #2

#4 Dihydroergotamine[Mesh] OR DHE*[tiab] OR D.H.E.*[tiab] OR Dihydroergo*[tiab] OR Ergotamine[Mesh] OR Ergotamin*[tiab]

#5 Serotonin 5-HT1 Receptor Agonists[Mesh] OR Serotonin 5 HT1 Receptor Agonist*[tiab] OR Serotonin 5-HT1B-D Agonist*[tiab] OR Serotonin 5 HT1B D Agonist*[tiab] OR 5-HT1B-D Agonist*[tiab] OR 5 HT1B D Agonist*[tiab]

#6 almotriptan[Supplementary Concept] OR almotriptan*[tiab]

#7 eletriptan[Supplementary Concept] OR UK 166,044[tiab] OR UK-166044[tiab] OR UK 166044[tiab] OR UK-166,044[tiab] OR UK-116044-04[tiab] OR UK-116,044-04[tiab] OR eletriptan*[tiab]

#8 frovatriptan[Supplementary Concept] OR VML-251[tiab] OR VML251[tiab] OR SB 209509[tiab] OR frovatriptan*[tiab]

#9 naratriptan[Supplementary Concept] OR GR 85548A[tiab] OR naratriptan*[tiab]

#10 rizatriptan[Supplementary Concept] OR rizatriptan*[tiab] OR MK 0462[tiab] OR MK-0462[tiab] OR MK-462[tiab] OR MK 462[tiab]

#11 Sumatriptan[Mesh] OR Sumatriptan*[tiab] OR GR-43175[tiab] OR GR 43175[tiab] OR GR43175[tiab]

#12 zolmitriptan[Supplementary Concept] OR zolmitriptan*[tiab] OR 311C90[tiab]

#13 lasmiditan[Supplementary Concept] OR lasmiditan*[tiab] OR COL-144[tiab] OR LY573144[tiab]

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#14 BMS-927711[tiab] OR rimegepant[tiab]
#15 ubrogepant[Supplementary Concept] OR MK-1602[tiab] OR ubrogepant[tiab]
#16 #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR e10 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15
#17 randomized controlled trial [pt]
#18 controlled clinical trial [pt]
#19 randomized [tiab]
#20 placebo [tiab]
#21 clinical trials as topic [mesh: noexp]
#22 randomly [tiab]
#23 trial [ti]
#24 #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23
#25 animals[mh] NOT humans[mh]
#26 #24 NOT #25
#27 #3 AND #16 AND #26

**EMBASE**

#1 ‘migraine’/exp OR ‘migrain*’:ab,ti,kw
#2 ‘drug therapy’/exp OR ‘drug*’:ab,ti,kw OR ‘pharma*’:ab,ti,kw or ‘chemotherap*’:ab,ti,kw OR ‘therap*’:ab,ti,kw OR ‘treat*’:ab,ti,kw OR 'treatment outcome'/exp OR ‘outcome*’:ab,ti,kw
#3 #1 AND #2
#4 ‘antimigraine agent’/exp
#6 ‘dihydroergo*’:ab,ti,kw OR ‘ergotamin*’:ab,ti,kw
#7 ‘almotriptan*’:ab,ti,kw

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#8 ‘eletriptan*’:ab,ti,kw OR ‘uk 116044*’:ab,ti,kw

#9 ‘frovatriptan*’:ab,ti,kw OR ‘vml 251’:ab,ti,kw OR ‘vml251’:ab,ti,kw

#10 ‘naratriptan*’:ab,ti,kw

#11 ‘mk 0462’:ab,ti,kw OR ‘mk 462’:ab,ti,kw OR ‘mk0462’:ab,ti,kw OR ‘mk462’:ab,ti,kw OR ‘rizatriptan*’:ab,ti,kw

#12 ‘sumatriptan*’:ab,ti,kw OR ‘GR 43175’:ab,ti,kw

#13 ‘zolmitriptan’:ab,ti,kw OR ‘311C90’:ab,ti,kw

#14 ‘col 144’:ab,ti,kw OR ‘col144’:ab,ti,kw OR ‘lasmiditan*’:ab,ti,kw OR ‘ly 573144’:ab,ti,kw OR ‘ly573144’:ab,ti,kw

#15 ‘bms 927711’:ab,ti,kw OR ‘bms927711’:ab,ti,kw OR ‘rimegepant*’:ab,ti,kw

#16 ‘mk 1602’:ab,ti,kw OR ‘mk1602’:ab,ti,kw OR ‘ubrogepant’:ab,ti,kw

#17 #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR e10 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16

#18 random*:ab,ti OR placebo*:de,ab,ti OR (double NEXT/1 blind*):ab,ti

#19 #3 AND #17 AND #18

**Cochrane Library**

#1 MeSH descriptor: [Migraine disorders] explode all trees

#2 (migraine*):ti,ab,kw

#3 #1 OR #2

#4 MeSH descriptor: [Ergotamine] explode all trees

#5 (ergotamin*):ti,ab,kw

#6 MeSH descriptor: [Dihydroergotamine] explode all trees

#7 (dihydroergotamin*):ti,ab,kw

#8 MeSH descriptor: [Serotonin 5 HT1 Receptor Agonists] explode all trees

#9 (Serotonin 5 HT1 Receptor Agonist*):ti,ab,kw

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| | |
|---|---|
| #10 | (almotriptan*):ti,ab,kw |
| #11 | (eletriptan*):ti,ab,kw |
| #12 | (frovatriptan*):ti,ab,kw |
| #13 | (naratriptan*):ti,ab,kw |
| #14 | (rizatriptan*):ti,ab,kw |
| #15 | MeSH descriptor: [Sumatriptan] explode all trees |
| #16 | (sumatriptan*):ti,ab,kw |
| #17 | (zolmitriptan*):ti,ab,kw |
| #18 | (lasmiditan*):ti,ab.kw |
| #19 | (rimegepant):ti,ab,kw |
| #20 | (ubrogepant):ti,ab,kw |
| #21 | #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR e10 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 |
| #22 | #3 AND #21 |

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**eTable 1.** Study Characteristics

| Study | Country | Proportion of female | Mean age (years) | Pre-treatment headache intensity | Aura | USAe of preventive medications |
|---|---|---|---|---|---|---|
| The Oral Sumatriptan Dose-Defining Study Group, 1991[1] | Sweden, UK, Austria, Belgium, Germany, Finland, France, and Netherlands | 83% | 40 | N.A. | With aura 66%<br>Without aura 34% | Not allowed |
| The Oral Sumatriptan International Multiple-Dose Study Group, 1991[2] | Sweden, UK, Finland, France, and Denmark | 85% | 41 | Moderate 62%<br>Severe 38% | With aura 33%<br>Without aura 67% | Not allowed |
| Giuseppe Nappi et al., 1994[3] | Italy | 77% | 38 | Moderate 49%<br>Severe 51% | With aura 13%<br>Without aura 87% | Not allowed |
| N. Cutler et al., 1995[4] | USA | 91% | 39 | N.A. | N.A. | Not allowed |
| Paul Winner et al., 1996[5] | USA | 88% | 41 | Moderate 51%<br>Severe 49% | N.A. | Allowed |

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|---|---|---|---|---|---|---|
| W. Hester Visser et al., 1996[6] | USA and Netherlands | 89% | 40 | Moderate 49%<br>Severe 51% | N.A. | Not allowed |
| The Dihydroergotamine Working Group, 1996[7] | USA | N.A. | 40 | N.A. | N.A. | Not allowed |
| H Gijsmant et al., 1997[8] | USA and Netherlands | 86% | 39 | Moderate 57%<br>Severe 43% | N.A. | Not allowed |
| Arthur Klassen et al., 1997[9] | USA | 87% | 40 | N.A. | With aura 6%<br>Without aura 71%<br>Mixed 22% | Not allowed |
| V. Pfaffenrath et al., 1998[10] | Germany, Netherland, France, Norway, Sweden, Italy, and the Czech Republic | 84% | 40 | N.A. | With aura 20%<br>Without aura 66%<br>Mixed 15% | Not allowed |
| Judith Teall et al., 1998[11] | USA & 9 other countries | 86% | 41 | Moderate 68%<br>Severe 32% | With aura 13% | Not allowed |
| SP Ahrens et al., 1999[12] | USA & 4 other countries | 88% | 42 | N.A. | With aura 14% | Not allowed |

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| | | | | | | |
|---|---|---|---|---|---|---|
| N Savani et al., 1995[13] | Belgium, Finland, Hungary, New Zealand, Norway, and UK | 86% | 39 | N.A. | N.A. | Not allowed |
| Martin Bomhof et al., 1999[14] | 46 international sites | 84% | 39 | Moderate 47%<br>Severe 53% | With aura 11%<br>Without aura 89% | Allowed |
| R. Michael Gallagher et al., 2000[15] | USA | 88% | 41 | N.A. | With aura 18%<br>Without aura 57%<br>Mixed 25% | Not allowed |
| J. Pascual et al., 2000[16] | 66 international sites | 83% | 39 | Moderate 47%<br>Severe 48%<br>Unknown 5% | With aura 12%<br>Without aura 87%<br>Unknown <1% | Allowed |
| J. Pascual et al., 2000[17] | Belgium, Czech Republic, Estonia, Germany, Hungary, Poland, Spain Netherlands, and UK | 87% | 41 | N.A. | N.A. | Not allowed |

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|---|---|---|---|---|---|---|
| H. Havanka et al., 2000[18] | Austria, Belgium, Denmark, Finland, France, Germany, Israel, Netherlands, New Zealand, Poland, Portugal, and Sweden | 88% | N.A. | Moderate 28%<br>Severe 72% | With aura 8%<br>Without aura 75%<br>Mixed 16% | Not allowed |
| P J Goadsby et al., 2000[19] | Europe, and Australia | 82% | 40 | Moderate 57%<br>Severe 43% | With aura 12%<br>Without aura 69%<br>Mixed 19% | Not allowed |

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|---|---|---|---|---|---|---|
| K Gruffyd-Jones et al., 2001[20] | Argentina, Austria, Belgium, Czech Republic, Denmark, Finland, Germany, Hungary, Mexico, Norway, Portugal, Poland, Sweden, Spain, South Africa, Switzerland, and UK | 85% | 42 | N.A. | With aura 16%<br>Without aura 57%<br>Mixed 28% | Not allowed |
| Egilius Spierings et al., 2001[21] | USA | 89% | 41 | Moderate 67%<br>Severe 33% | N.A. | Not allowed |
| C. Dahlöf et al., 2001[22] | Portugal, Sweden, United Kingdom, Hungary, Estonia, France, Germany, Poland, the Netherlands, and Denmark | 85% | 41 | Moderate 58%<br>Severe 42% | With aura 21%<br>Without aura 79% | Not allowed |

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|---|---|---|---|---|---|---|
| R. Stark et al., 2002[23] | Australia, Europe, and South Africa | 83% | 42 | Moderate 60%<br>Severe 40% | With aura 8%<br>Without aura 65%<br>Mixed 27% | Not allowed |
| AJ Dowson et al., 2002[24] | Canada, South Africa, and<br>UK | 87% | 42 | Moderate 71%<br>Severe 29% | With aura 24% | Not allowed |
| Alan Rapoport et al., 2002[25] | USA | 86% | 41 | Moderate 66%<br>Severe 34% | With aura 15%<br>Without aura 63%<br>Mixed 22% | Not allowed |
| F Sakai et al., 2002[26] | Japan | 74% | 38 | N.A. | With aura 36%<br>Without aura 64% | Not allowed |
| G. Sandrini et al., 2002[27] | Europe, Canada, and South Africa | 88% | 38 | N.A. | With aura 12%<br>Without aura 65%<br>Mixed 23% | Not allowed |
| Ninan T. Mathew et al., 2003[28] | 166 international sites | 87% | 42 | N.A. | With aura 15%<br>Without aura 65%<br>Mixed 20% | Not allowed |
| Fred Sheftell et al., 2002[29] | USA | 87% | 42 | Moderate 74%<br>Severe 26% | With aura 9%<br>Without aura 65%<br>Mixed 26% | Not allowed |

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|---|---|---|---|---|---|---|
| M Färkkilä et al., 2003[30] | Norway, Sweden, Finland, Denmark, and Netherlands | 80% | 41 | Moderate 69%<br>Severe 31% | With aura 12%<br>Without aura 59%<br>Mixed 29% | Not allowed |
| Robert Nett et al., 2003[31] | USA, Canada, and Puerto Rico | 100% | 36 | N.A. | With aura 8%<br>Without aura 75%<br>Mixed 17% | Not allowed |
| G Garcia-Ramos et al., 2003[32] | UK, and Latin America | 81% | 37 | N.A. | With aura 18%<br>Without aura 67%<br>Mixed 15% | Not allowed |

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| | | | | | | |
|---|---|---|---|---|---|---|
| T J Steiner et al., 2003[33] | Austria, Belgium, Croatia, Czech Republic, Denmark, France, Germany, Greece, Hungary, Italy, Netherlands, Norway, Poland, Portugal, Slovakia, Slovenia, Spain, and Switzerland | 85% | 40 | Moderate and Severe 85% | With aura 8%<br>Without aura 73%<br>Mixed 19% | Not allowed |
| S Landy et al., 2004[34] | Europe, Canada, and New Zealand | 100% | 38 | Moderate 30%<br>Severe 9% | N.A. | Not allowed |
| S Landy et al., 2004[34] | USA, Canada, and Puerto Rico | 100% | 37 | Moderate 6%<br>Severe 2% | N.A. | Not allowed |

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|---|---|---|---|---|---|---|
| Egilius Spierings et al., 2004[35] | N.A. | 87% | 42 | N.A. | With aura 12%<br>Without aura 66%<br>Mixed 22% | Not allowed |
| Elizabeth Loder et al., 2005[36] | USA | 83% | 41 | N.A. | With aura 9%<br>Without aura 72%<br>Mixed 19% | Not allowed |
| Hans-Christoph Diener et al., 2005[37] | Germany | 87% | 41 | Moderate 29%<br>Severe 71% | N.A. | Not allowed |
| J L Brandes et al., 2005[38] | USA and Canada | 82% | 39 | N.A. | Without aura 69% | Not allowed |
| H Massiou et al., 2005[39] | France | 100% | N.A. | Moderate 49%<br>Severe 21% | With aura 19%<br>Without aura 70%<br>Mixed 11% | Not allowed |
| Susan E Jelinski et al., 2006[40] | Canada | 85% | 41 | N.A. | With aura 9%<br>Without aura 67%<br>Mixed 24% | Not allowed |
| Roger Cady et al., 2006[41] | USA | 89% | 43 | N.A. | N.A. | Not allowed |
| Roger Cady et al., 2006[41] | USA | 88% | 41 | N.A. | N.A. | Not allowed |

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|---|---|---|---|---|---|---|
| Peer Tfelt-Hansen et al., 2006[42] | N.A. | 78% | 38 | N.A. | With aura 11%<br>Without aura 80%<br>Mixed 9% | Allowed |
| Michael Tuchman et al., 2006[43] | USA | 100% | 38 | N.A. | With aura 8%<br>Without aura 73%<br>Mixed 19% | Not allowed |
| P J Goadsby et al., 2007[44] | Belgium, Finland, France, Germany, Italy, Portugal, Spain, Sweden, and UK | 85% | 40 | Moderate 62%<br>Severe 38% | N.A. | Not allowed |
| Ninan T Mathew et al., 2007[45] | USA | 87% | 40 | N.A. | N.A. | Not allowed |
| Eletriptan Steering Committee in Japan et al., 2002[46] | Japan | 74% | 36 | Moderate 77%<br>Severe 23% | N.A. | Not allowed |

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| | | | | | | |
|---|---|---|---|---|---|---|
| P J Goadsby et al., 2008[47] | Belgium, France, Germany, Italy, and Portugal | 83% | 39 | N.A. | N.A. | Not allowed |
| Michel D Ferrari et al., 2010[48] | Germany, Finland, and Netherlands | 85% | 38 | N.A. | N.A. | Not allowed |
| Sheena K. Aurora et al., 2011[49] | USA | 92% | 42 | Moderate 54%<br>Severe 46% | N.A. | Not allowed |
| Markus Färkkilä et al., 2012[50] | Belgium, France, Finland, Germany, and Spain | 88% | 40 | Moderate 58%<br>Severe 42% | With aura 8%<br>Without aura 91% | Not allowed |
| Piero Barbanti et al., 2012[51] | N.A. | 83% | 43 | Moderate 69%<br>Severe 31% | N.A. | Not allowed |
| Jerome Goldstein et al., 2012[52] | USA | 85% | 41 | Moderate 77%<br>Severe 23% | N.A. | Not allowed |
| Tiffini Voss et al., 2016[53] | USA | 87% | 41 | Moderate 67%<br>Severe 33% | With aura 30.2%<br>Without aura 69.4% | Not allowed |
| Egilius Spierings et al., 2018[54] | USA | 87% | 41 | Moderate 49%<br>Severe 51% | N.A. | Not allowed |
| Richard B Lipton et | USA | 81% | 42 | N.A. | N.A. | Not allowed |

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|---|---|---|---|---|---|---|
| al., 2018[55] | | | | | | |
| Richard B Lipton et al., 2018[56] | USA | 86% | 42 | N.A. | N.A. | Allowed |
| Bernice Kuca et al., 2018[57] | USA | 84% | 42 | Moderate 28%<br>Severe 72% | N.A. | Not allowed |
| Richard B Lipton et al., 2019[58] | N.A. | 89% | 41 | N.A. | N.A. | Not allowed |
| Peter J Goadsby et al., 2019[59] | USA, UK, and Germany | 84% | 43 | Moderate 70%<br>Severe 30% | With aura 36%<br>Without aura 63% | Not allowed |
| Robert Croop et al., 2019[60] | USA | 85% | 40 | N.A. | With aura 30%<br>Without aura 70% | Not allowed |
| Richard B Lipton et al., 2019[61] | N.A. | 90% | 42 | Moderate 59%<br>Severe 41% | With aura 51%<br>Without aura 24%<br>Mixed 25% | Not allowed |
| David W Dodick et al., 2019[62] | USA | 88% | 41 | Moderate 63%<br>Severe 37% | N.A. | Not allowed |

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**eTable 2.** SUCRA of Outcomes for Anti-Migraine Specific Treatments in Dosages With Widespread Clinical Use

| Treatment* | Pain freedom | Pain relief | Adverse events |
|---|---|---|---|
| Eletriptan 40mg | 95.4 | 90.7 | 44.4 |
| Rizatriptan 10mg | 89.3 | 92.6 | 69.7 |
| Sumatriptan 100mg | 83.5 | 72.7 | 65 |
| Frovatriptan 2.5mg | 77.6 | 30.5 | 50.5 |
| Rizatriptan 5mg | 76.2 | 66.1 | 48 |
| Sumatriptan 50mg | 68 | 78.7 | 64.6 |
| Zolmitriptan 2.5mg | 66.7 | 67.3 | 80.6 |
| Eletriptan 20mg | 57.4 | 55.7 | 36.2 |
| Almotriptan 12.5mg | 53.5 | 58.7 | 32.9 |
| Sumatriptan 10mg nasal spray | 49.9 | 72.1 | 91.2 |
| Ubrogepant 100mg | 40.1 | 15.8 | 50.4 |
| Lasmiditan 100mg | 28.9 | 32.7 | 95.1 |
| Ubrogepant 50mg | 24.9 | 14 | 16.7 |
| Almotriptan 6.25mg | 23.8 | 17.1 | 16.3 |
| Rimegepant 75mg | 23.6 | 44.5 | 40.1 |
| Naratriptan 2.5mg | 22.2 | 37 | 25.9 |
| Lasmiditan 50mg | 18.4 | 11.5 | 88.8 |
| Dihydroergotamine 2mg nasal spray | N.A. | 95.5 | N.A. |
| Dihydroergotamine 3mg nasal spray | N.A. | 59.1 | N.A. |
| Naratriptan 1mg | N.A. | 37.9 | 14.9 |

*Sorted by order of mean rank for pain freedom (e.g., the former the greater possibility to be the best treatment in terms of pain freedom at hours).

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**eTable 3.** The Percentage of Serious Adverse Events in All Adverse Events With Respect to Specific Anti-Migraine Treatment

| Medication | Adverse events | The percentage of serious adverse events |
|---|---|---|
| Placebo | 1296 | 0.5% |
| Ergot alkaloids | | |
| Dihydroergotamine 1mg inhaler | 126 | 0.0% |
| Triptans | | |
| Almotriptan 6.25mg | 47 | 2.1% |
| Almotriptan 12.5mg | 196 | 1.0% |
| Eletriptan 20mg | 13 | 0.0% |
| Eletriptan 40mg | 402 | 0.0% |
| Eletriptan 80mg | 203 | 1.0% |
| Naratriptan 2.5mg | 62 | 0.0% |
| Rizatriptan 10mg | 324 | 0.3% |
| Sumatriptan 50mg | 304 | 0.0% |
| Sumatriptan 100mg | 427 | 0.5% |
| Sumatriptan 200mg | 139 | 0.7% |
| Sumatriptan 300mg | 164 | 0.6% |
| Zolmitriptan 1mg | 34 | 0.0% |
| Zolmitriptan 1.9mg | 37 | 0.0% |
| Zolmitriptan 3.8mg | 43 | 0.0% |
| Zolmitriptan 2.5mg | 733 | 0.5% |
| Zolmitriptan 5mg | 360 | 0.3% |
| Sumatriptan 10mg nasal spray | 5 | 0.0% |
| Sumatriptan 6.5mg transdermal system | 121 | 0.0% |
| Lasmiditan | | |
| Lasmiditan 50mg | 53 | 0.0% |
| Lasmiditan 100mg | 288 | 0.0% |
| Lasmiditan 200mg | 321 | 0.9% |
| Lasmiditan 400mg | 59 | 0.0% |
| CGRP antagonists | | |
| Rimegepant 75mg | 252 | 1.2% |
| Ubrogepant 1mg | 33 | 0.0% |
| Ubrogepant 10mg | 29 | 0.0% |
| Ubrogepant 25mg | 65 | 0.0% |

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| | | |
|---|---|---|
| Ubrogepant 50mg | 130 | 0.8% |
| Ubrogepant 100mg | 109 | 0.0% |

CGRP, calcitonin gene-related peptide.

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**eTable 4.** The Percentage of Chest Symptoms in All Adverse Events With Respect to Specific Anti-Migraine Treatment

| Medication | Adverse events | The percentage of chest symptoms |
|---|---|---|
| Placebo | 661 | 2.0% |
| Triptans | | |
| Almotriptan 2mg | 29 | 0.0% |
| Almotriptan 6.25mg | 27 | 3.7% |
| Almotriptan 12.5mg | 167 | 1.8% |
| Almotriptan 25mg | 41 | 7.3% |
| Eletriptan 40mg | 376 | 5.9% |
| Eletriptan 80mg | 168 | 7.7% |
| Frovatriptan 0.5mg | 39 | 2.6% |
| Frovatriptan 1mg | 37 | 0.0% |
| Frovatriptan 2.5mg | 90 | 2.2% |
| Frovatriptan 5mg | 95 | 7.4% |
| Frovatriptan 10mg | 114 | 8.8% |
| Frovatriptan 20mg | 137 | 12.4% |
| Frovatriptan 40mg | 147 | 14.3% |
| Naratriptan 2.5mg | 73 | 6.8% |
| Rizatriptan 5mg | 139 | 5.0% |
| Rizatriptan 10mg | 371 | 7.0% |
| Rizatriptan 20mg | 31 | 6.5% |
| Rizatriptan 40mg | 66 | 9.1% |
| Sumatriptan 25mg | 74 | 4.1% |
| Sumatriptan 50mg | 299 | 8.4% |
| Sumatriptan 100mg | 609 | 6.7% |
| Sumatriptan 200mg | 139 | 2.2% |
| Sumatriptan 300mg | 164 | 9.8% |
| Sumatriptan 10mg nasal spray | 5 | 20.0% |
| Zolmitriptan 2.5mg | 366 | 9.0% |
| CGRP antagonists | | |
| Ubrogepant 1mg | 33 | 0.0% |
| Ubrogepant 10mg | 29 | 0.0% |
| Ubrogepant 25mg | 21 | 0.0% |
| Ubrogepant 50mg | 23 | 4.3% |

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| | | |
|---|---|---|
| Ubrogepant 100mg | 30 | 0.0% |

CGRP, calcitonin gene-related peptide.

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**eTable 5.** Significant Loop-Specific Inconsistencies of Network Meta-Analysis

| Loop | Ratio of odds ratios | 95% confidence interval | $Tau^2$ |
|---|---|---|---|
| **Pain relief** | | | |
| Placebo-zolmitriptan 2.5mg-zolmitriptan 5mg | 3.79 | (2.08, 6.90) | 0.026 |
| Placebo-sumatriptan 50mg-zomitriptan 2.5mg | 1.66 | (1.01, 2.70) | 0.020 |

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**eTable 6.** Significant Side-Splitting Inconsistencies of Network Meta-Analysis

| | Direct | | Indirect | | Difference | | Tau |
|---|---|---|---|---|---|---|---|
| Side | Coefficient | SE | Coefficient | SE | Coefficient | SE | |
| **Pain freedom** | | | | | | | |
| Symmetric | | | | | | | |
| Eletriptan 20mg vs. placebo | 0.90 | 0.18 | 1.77 | 0.32 | -0.87 | 0.39 | 0.15 |
| Sumatriptan 100mg vs. eletriptan 40mg | -0.45 | 0.14 | -0.02 | 0.16 | -0.43 | 0.22 | 0.15 |
| Asymmetric | | | | | | | |
| Sumatriptan 50mg vs. placebo | 1.08 | 0.12 | 1.52 | 0.15 | -0.44 | 0.19 | 0.14 |
| **Pain relief** | | | | | | | |
| Symmetric | | | | | | | |
| Zolmitriptan 2.5mg vs. placebo | 1.53 | 0.13 | 0.98 | 0.13 | 0.55 | 0.19 | 0.16 |
| Zolmitriptan 5mg vs. placebo | 0.28 | 0.18 | 1.28 | 0.13 | -1.00 | 0.22 | 0.11 |
| Zolmitriptan 5mg vs. zolmitriptan 2.5mg | 0.05 | 0.11 | -1.23 | 0.17 | 1.27 | 0.21 | 0.07 |
| Lasmiditan 400mg vs. lasmiditan 100mg | 0.02 | 0.39 | 1.43 | 0.57 | -1.41 | 0.68 | 0.18 |
| Asymmetric | | | | | | | |
| Zolmitriptan 2.5mg vs. placebo | 1.42 | 0.12 | 1.00 | 0.14 | 0.42 | 0.18 | 0.17 |
| Zolmitriptan 5mg vs. placebo | 0.41 | 0.18 | 1.22 | 0.14 | -0.81 | 0.23 | 0.12 |
| Zolmitriptan 2.5mg vs. sumatriptan 50mg | -0.15 | 0.14 | 0.37 | 0.14 | -0.52 | 0.19 | 0.16 |

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| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Zolmitriptan 5mg vs. sumatriptan 50mg | 0.00 | 0.13 | -0.81 | 0.21 | 0.81 | 0.23 | 0.12 |
| Zolmitriptan 5mg vs. zolmitriptan 2.5mg | -0.08 | 0.11 | -1.25 | 0.20 | 1.17 | 0.22 | 0.10 |
| **Adverse events** | | | | | | | |
| Symmetric | | | | | | | |
| Eletriptan 20mg vs. placebo | 0.62 | 0.31 | -0.99 | 0.47 | 1.61 | 0.58 | 0.26 |
| Eletriptan 40mg vs. placebo | 0.10 | 0.16 | 0.95 | 0.34 | -0.85 | 0.39 | 0.26 |
| Lasmiditan 400mg vs. placebo | 2.94 | 0.50 | 0.64 | 0.73 | 2.30 | 0.84 | 0.27 |
| Lasmiditan 400mg vs. lasmiditan 200mg | -0.13 | 0.55 | 1.79 | 0.76 | -1.92 | | |
| Asymmetric | | | | | | | |
| Zolmitriptan 2.5mg vs. eletriptan 40mg | 0.06 | 0.27 | 0.75 | 0.20 | -0.69 | 0.30 | 0.27 |
| Zolmitriptan 2.5mg vs. eletriptan 80mg | -0.60 | 0.28 | 0.09 | 0.23 | -0.69 | 0.30 | 0.27 |

SE, standard error

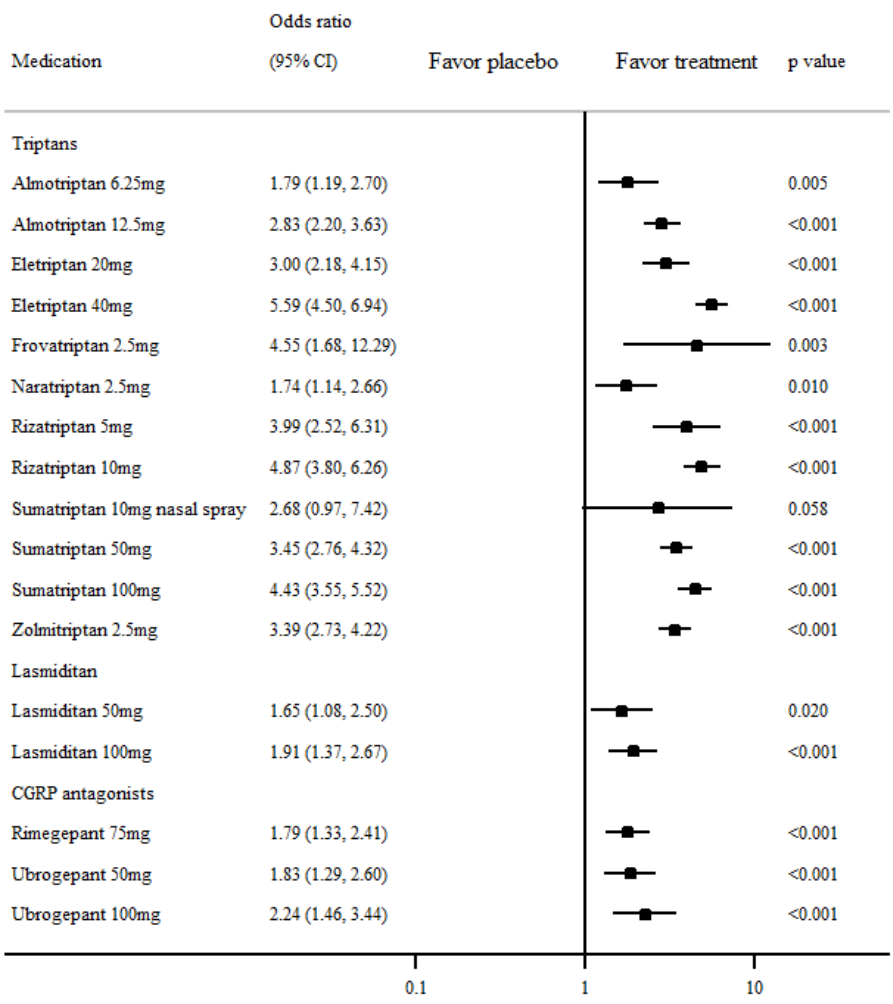
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**eTable 7.** Design-by-Treatment Interaction Model for Inconsistency of Network Meta-Analysis

| | $Chi^2$ | $Prob>Chi^2$ |
|---|---|---|
| Pain freedom | 62.65 | 0.10 |
| Pain relief | 61.01 | 0.24 |
| Adverse events | 50.53 | 0.30 |

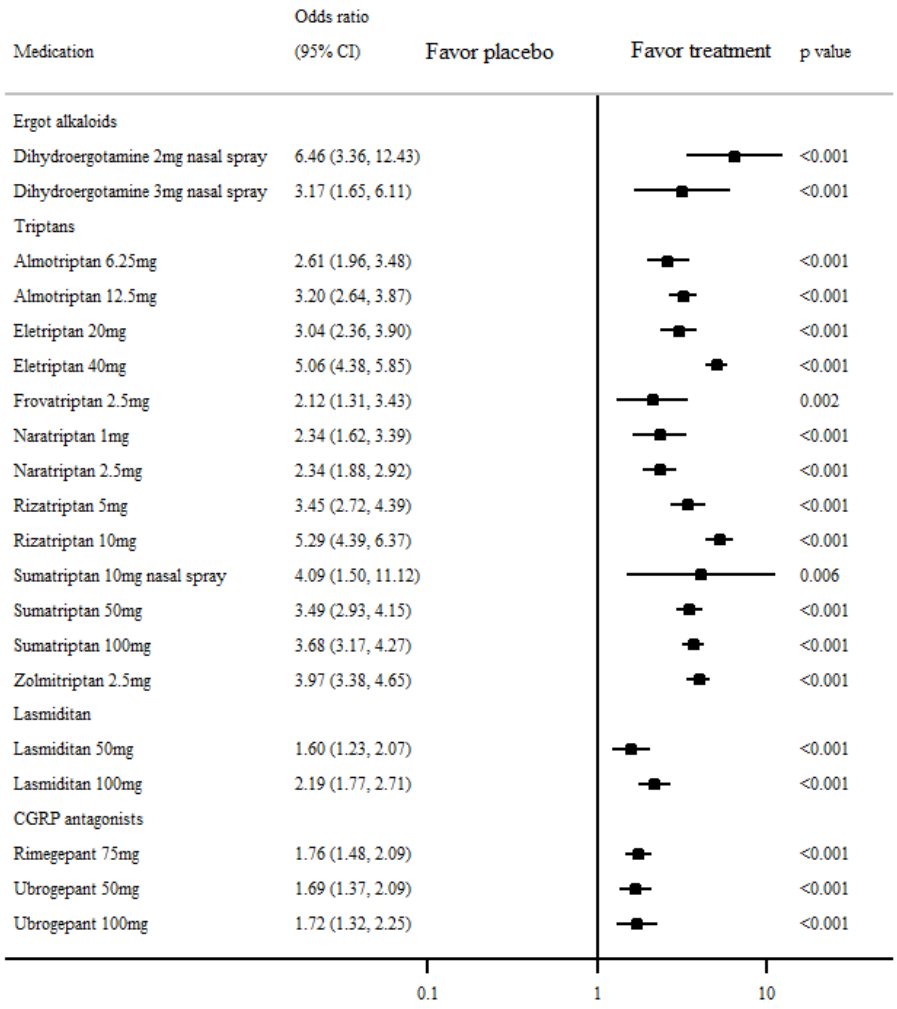
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**eFigure 1A.** Forest Plot of the Primary Outcome: Pain Freedom at 2 Hours for Specific Anti-Migraine Treatments in Dosages With Widespread Clinical Use



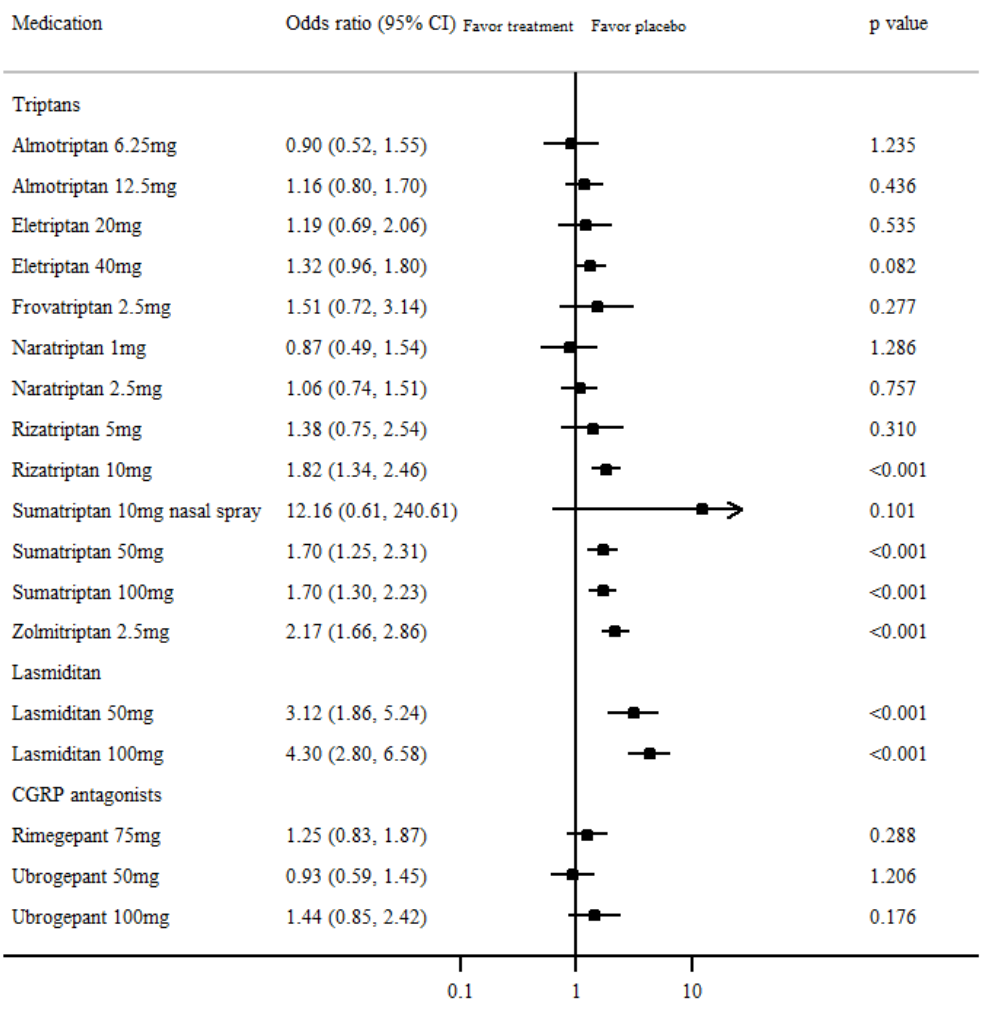

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**eFigure 1B.** Forest Plot of the Secondary Outcome: Pain Relief at 2 Hours for Specific Anti-Migraine Treatments in Dosages With Widespread Clinical Use



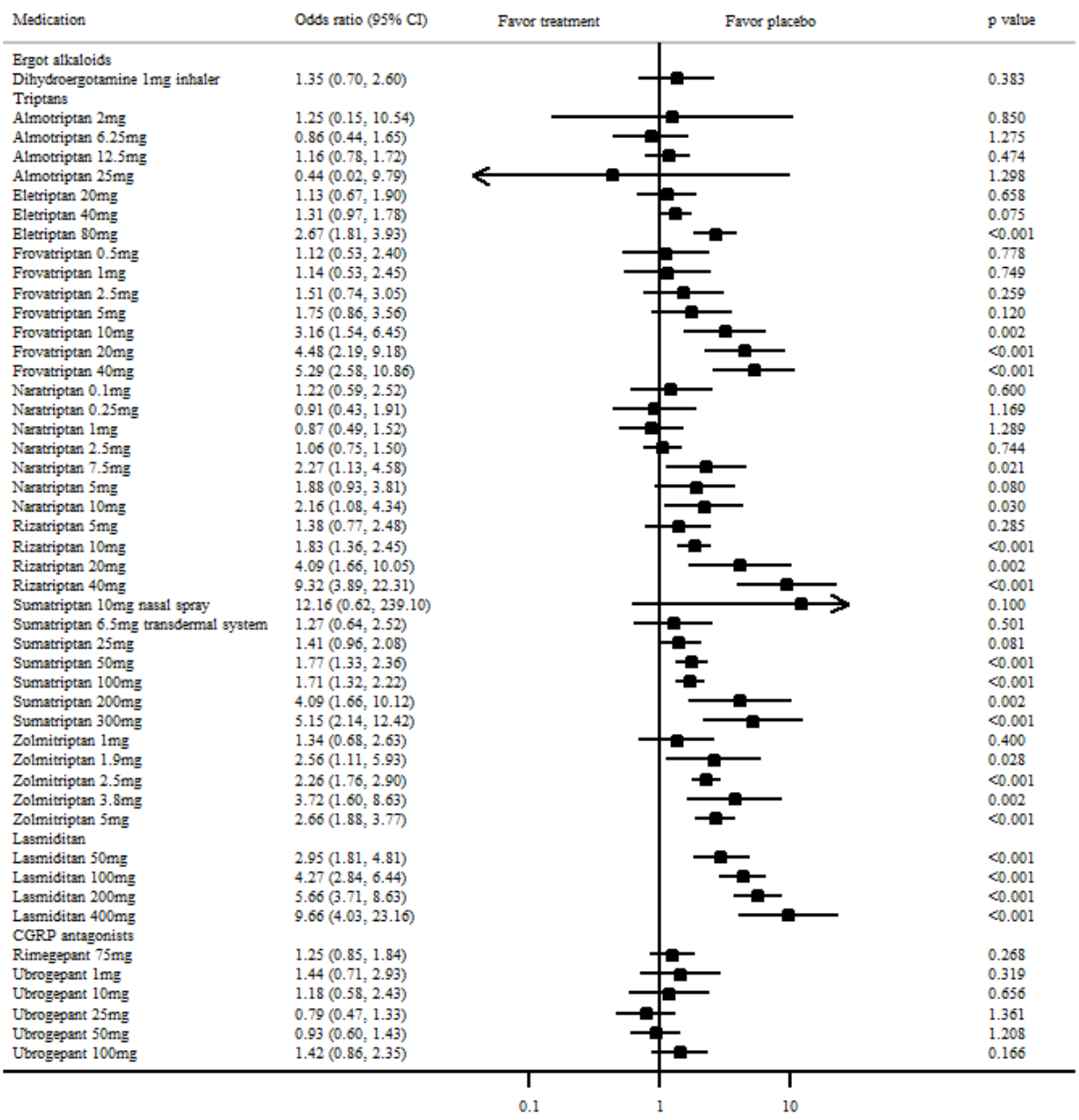

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**eFigure 1C.** Forest Plot of the Secondary Outcome: Adverse Events for Specific Anti-Migraine Treatments in Dosages With Widespread Clinical Use



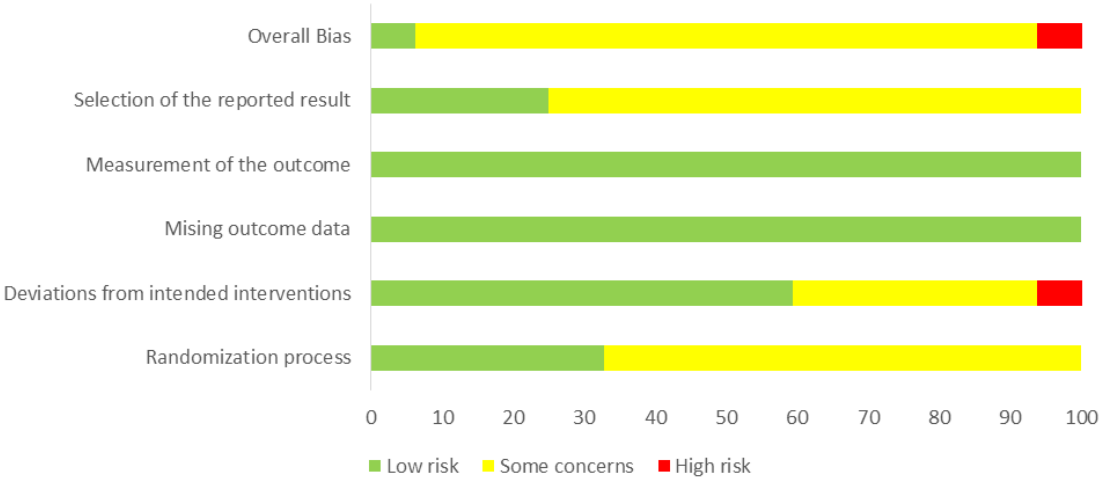

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**eFigure 2.** Forest Plot of the Secondary Outcome: Adverse Events



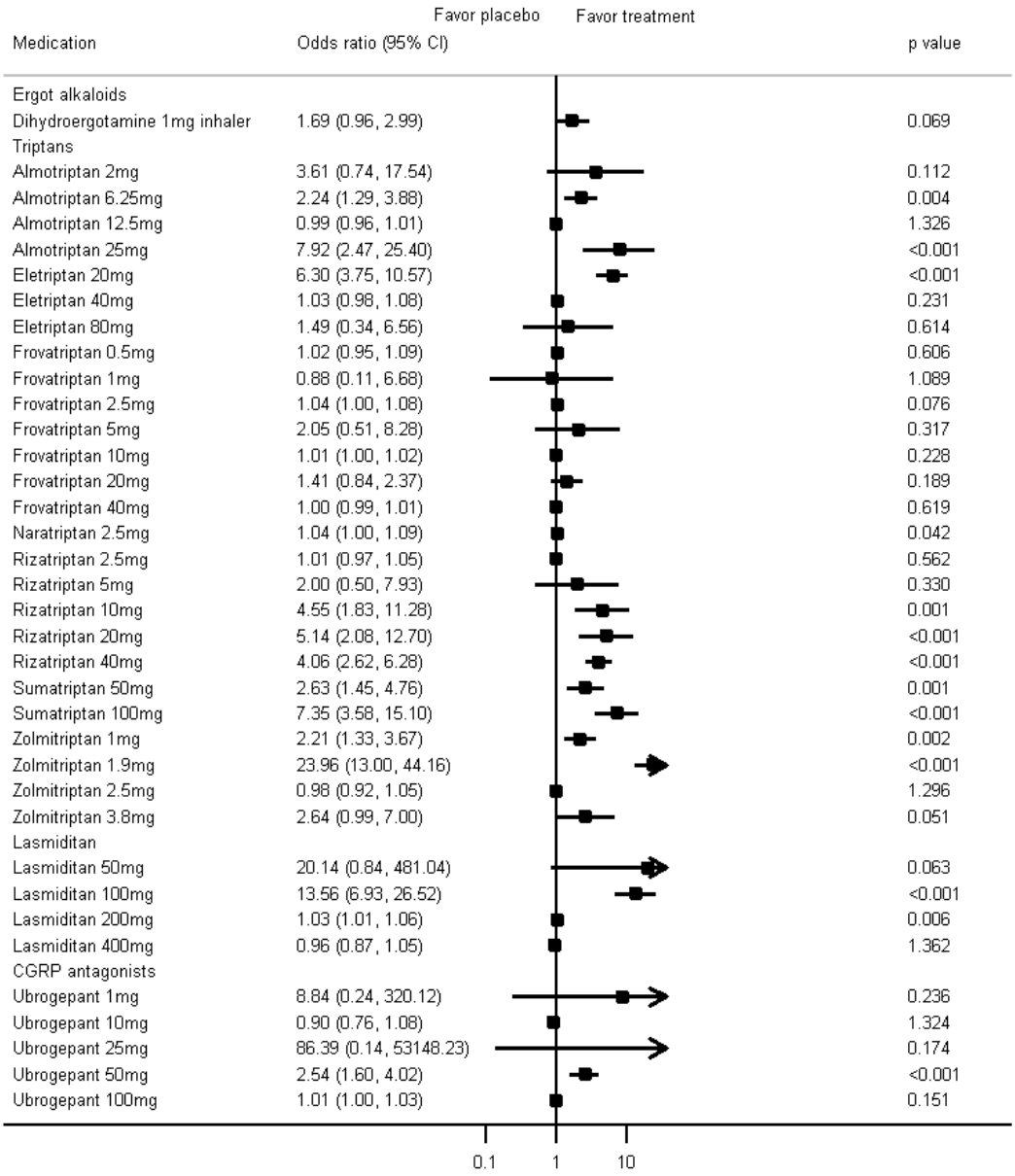

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**eFigure 3.** Risk of Bias



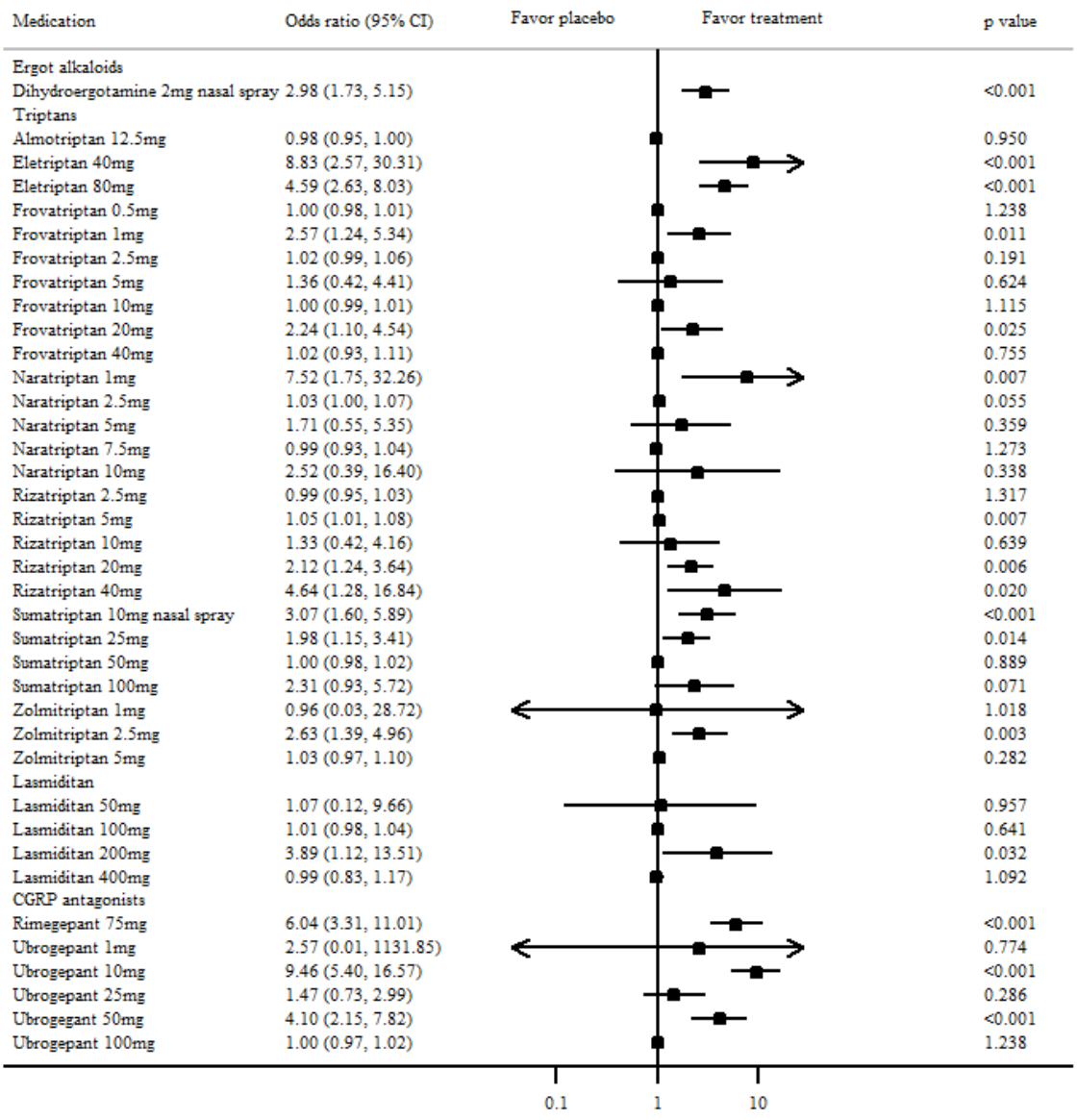

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**eFigure 4A.** Forest Plot of the Primary Outcome: Pain Freedom at 2 Hours for Specific Anti-Migraine Treatments After Adjustment for the Percentage of Severe Headache at Baseline



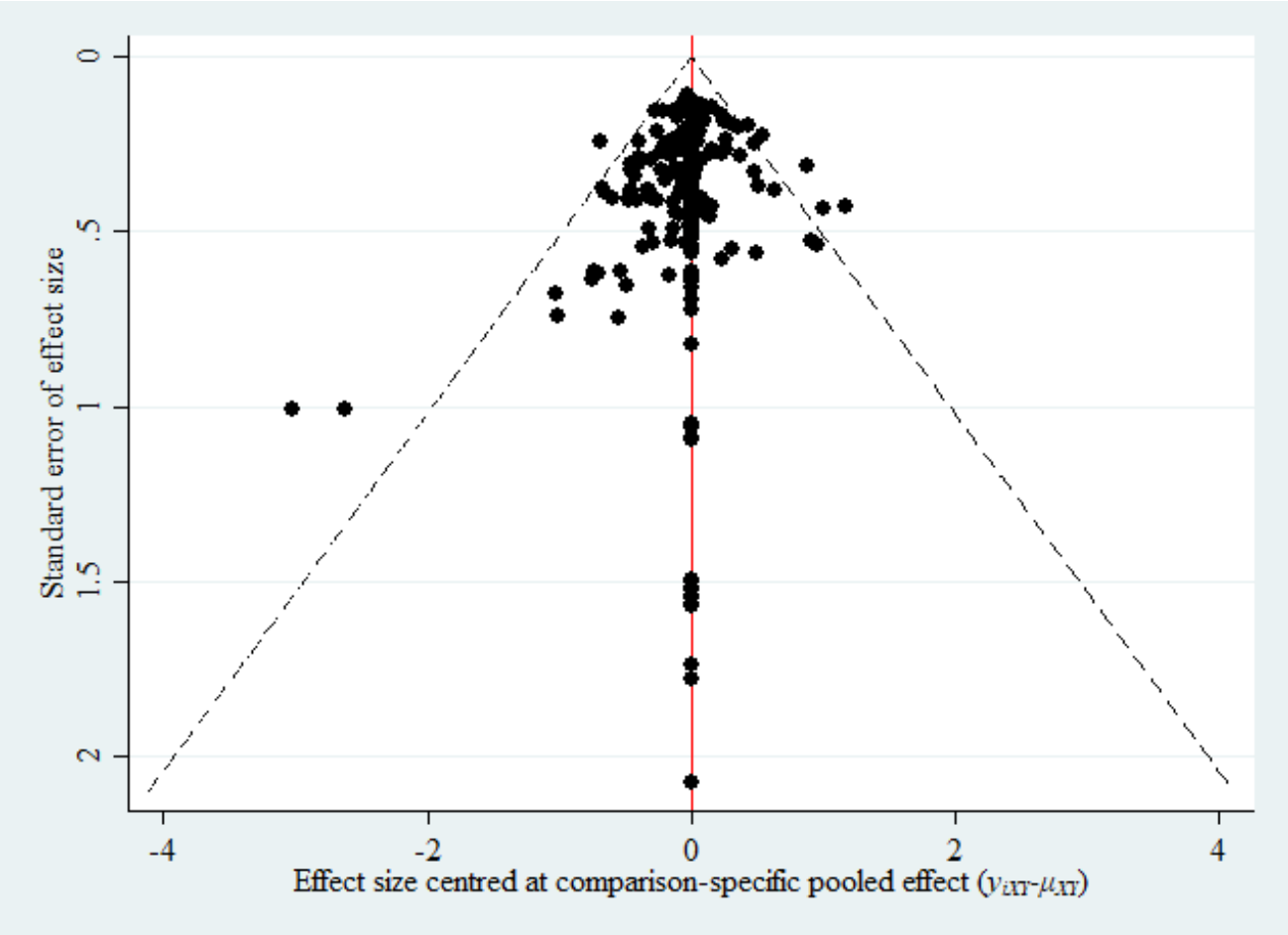

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**eFigure 4B.** Forest Plot of the Secondary Outcome: Pain Relief at 2 Hours for Specific Anti-Migraine Treatments After Adjustment for the Percentage of Severe Headache at Baseline



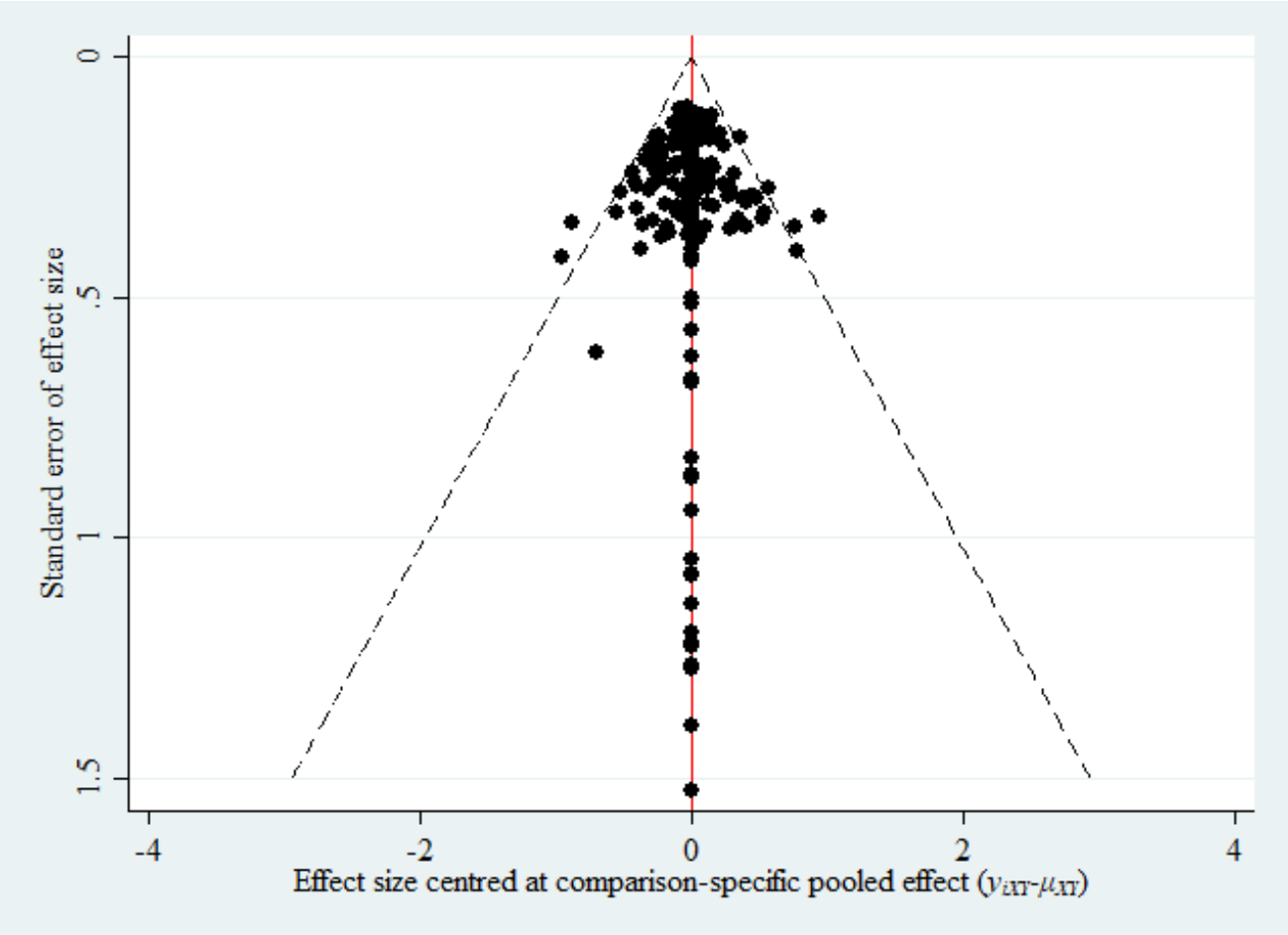

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**eFigure 5A.** Funnel Plot of the Primary Outcome: Pain Freedom at 2 Hours for Specific Anti-Migraine Treatments



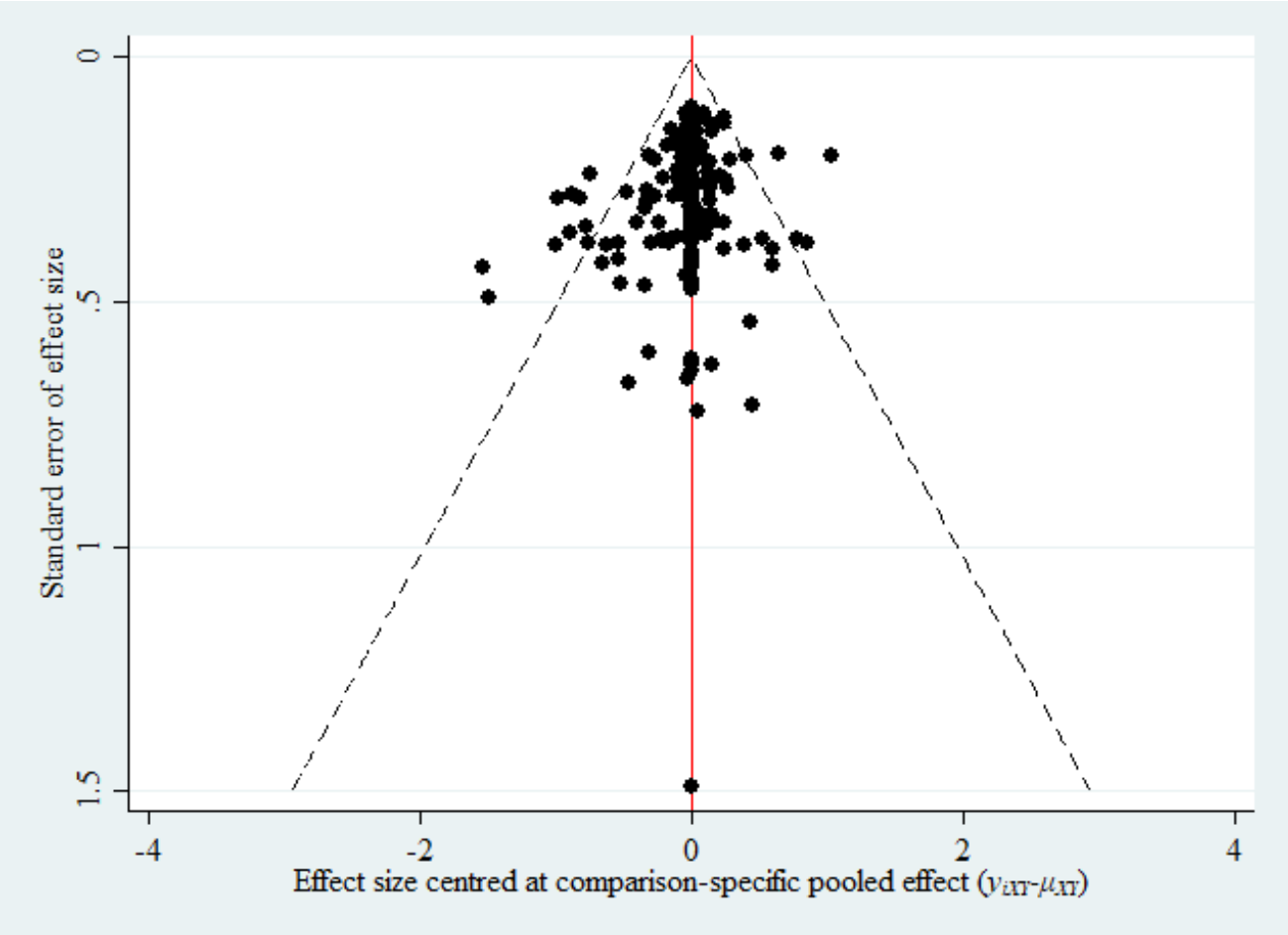

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**eFigure 5B.** Funnel Plot of the Secondary Outcome: Pain Relief at 2 Hours for Specific Anti-Migraine Treatments



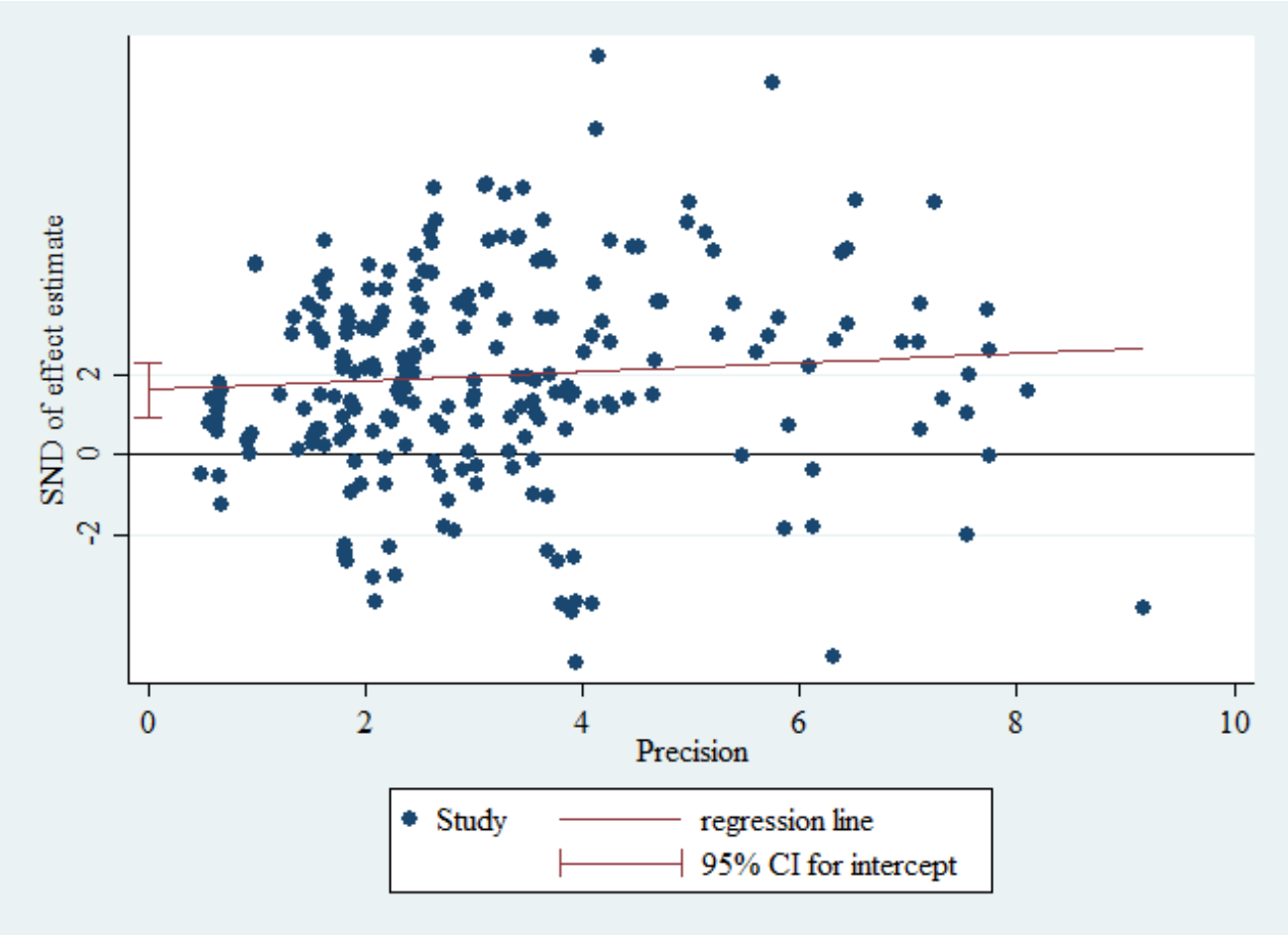

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**eFigure 5C.** Funnel Plot of the Secondary Outcome: Adverse Events for Specific Anti-Migraine Treatments



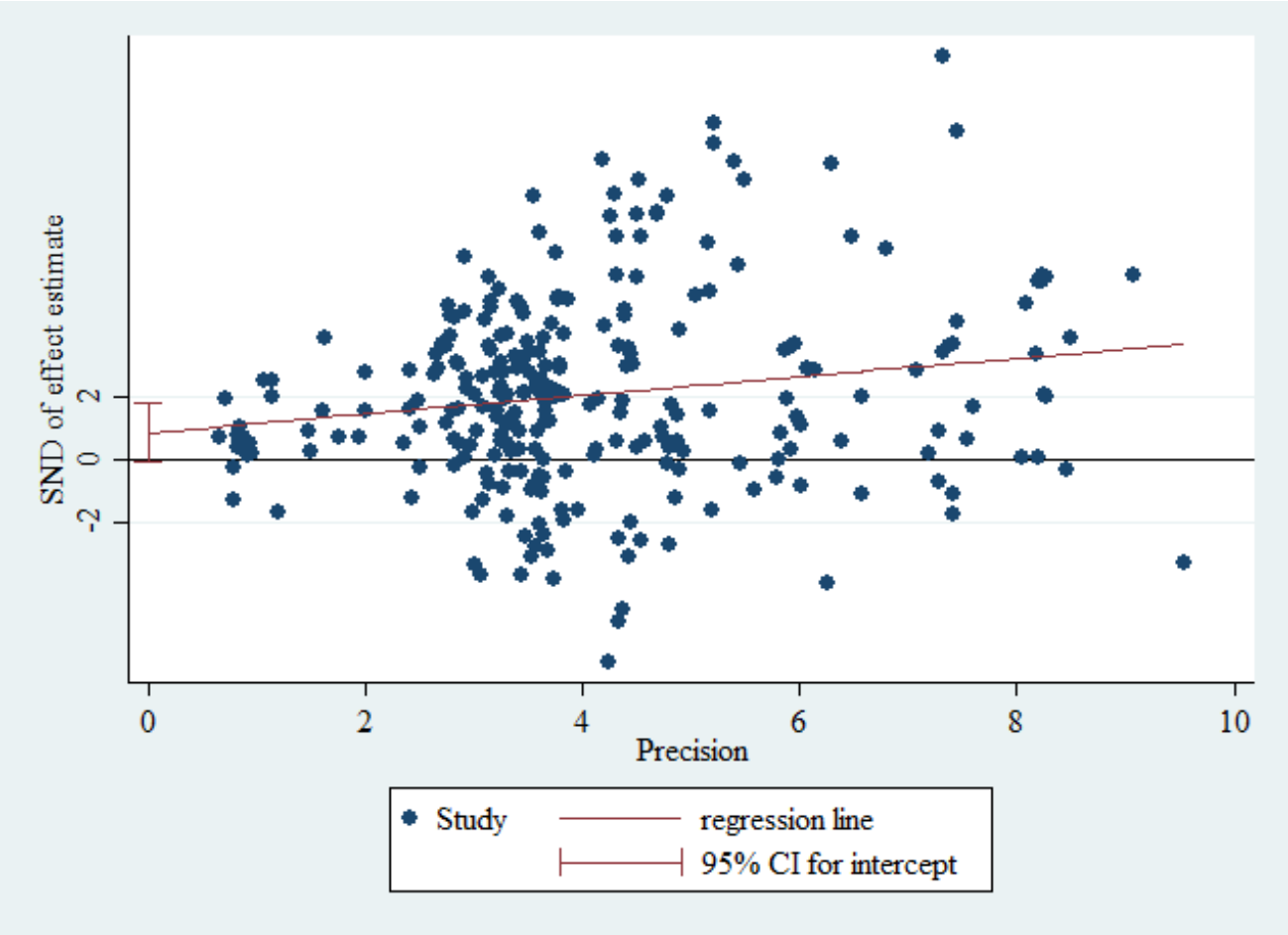

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**eFigure 6A.** Egger's Regression of All Specific Anti-Migraine Treatments for the Primary Outcome: Pain Freedom at 2 Hours



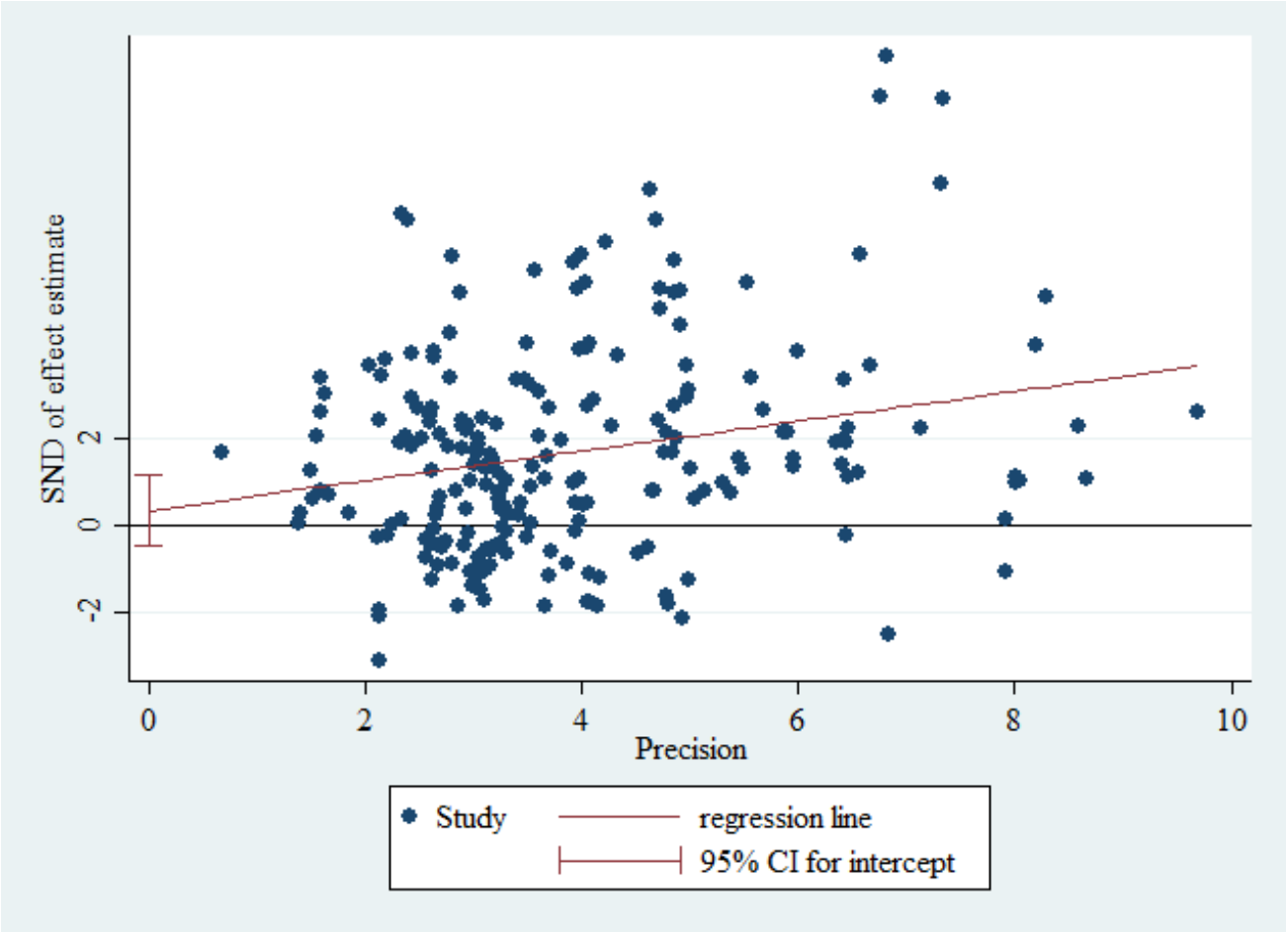

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**eFigure 6B.** Egger's Regression of All Specific Anti-Migraine Treatments for the Secondary Outcome: Pain Relief at 2 Hours




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**eFigure 6C.** Egger's Regression of All Specific Anti-Migraine Treatments for the Secondary Outcome: Adverse Events




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